Heaven's Light is Our Guide

Department of Electronics & Telecommunication Engineering

Rajshahi University of Engineering & Technology

WattGaurd - Automatic Power Factor Correction & Voltage Protection System

Author

Bhismodev Saha Dhairjo

Roll No. 2204012

Supervised by

Rifa Tabassum Mim

Lecturer

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Acknowledgement

I would like to express my heartfelt gratitude to everyone who supported me throughout the completion of my project titled **“WattGuard – Automatic Power Factor Correction and Voltage Protection System.”**

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05/11/2025 Bhismodev Saha Dhairjo

RUET, Rajshahi-6204, Bangladesh

Declaration

I hereby declare that this project report titled **"WattGaurd - Automatic Power Factor Correction & Voltage Protection System"** is my own work and, to the best of my knowledge, does not contain any material previously published or written by another person, nor material that has been submitted for the award of any degree or diploma at Department of Electronics & Telecommunication Engineering, Rajshahi University of Engineering & Technology, or any other institution, except where proper acknowledgment has been made. Contributions from others, including fellow students and AI tools, have been fully acknowledged. The intellectual content of this report is the result of my own effort, except where assistance from others has been explicitly acknowledged.

Author

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(**Bhismodev Saha Dhairjo**)
Roll No. 2204012
Department of Electronics & Telecommunication Engineering
Rajshahi University of Engineering & Technology
Rajshahi-6204, Bangladesh.

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Certificate

*This is to certify that the project entitled "**WattGaurd - Automatic Power Factor Correction & Voltage Protection System**" has been carried out by **Bhismodev Saha Dhairjo** under the supervision of **Rifa Tabassum Mim**, Lecturer, Department of Electronics & Telecommunication Engineering, Rajshahi University of Engineering & Technology.*

Supervisor

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(**Rifa Tabassum Mim**)

Lecturer

Department of Electronics & Telecommunication Engineering

Rajshahi University of Engineering & Technology

Rajshahi-6204, Bangladesh.

Abstract

This report presents **WattGuard**, an integrated protection and monitoring system designed to enhance electrical safety and power efficiency through automatic power factor correction and real-time energy analysis. Voltage fluctuations, including under- and over-voltage conditions, pose severe risks to electrical equipment, leading to premature failure and energy inefficiency. The proposed system addresses these challenges by combining a voltage protection mechanism with continuous monitoring of key electrical parameters such as voltage, current, energy consumption, and power factor.

The system employs an Arduino Uno microcontroller as the central processing unit, integrated with transformer-based voltage sensing, current sensors (ACS712 and SCT-013), and an LM358-based zero-crossing detection circuit for accurate phase angle and power factor estimation. Experimental evaluation under various load conditions validates the system's capability to detect voltage anomalies, improve power factor, and provide real-time parameter visualization. By unifying protection and efficiency in a single platform, WattGuard offers a cost-effective and scalable solution suitable for both residential and industrial power systems.

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Chapter 1
Introduction

This chapter introduces the project titled **"WattGuard – Automatic Power Factor Correction & Voltage Protection System."** The primary objective of this work is to design and implement an integrated system that enhances electrical safety by detecting under-voltage and over-voltage conditions while simultaneously measuring and monitoring essential electrical parameters such as energy consumption and power factor. Voltage fluctuations and poor power factor not only reduce the efficiency of electrical systems but also shorten the operational lifespan of connected appliances. Hence, developing a system that provides both protection against voltage anomalies and effective energy optimization is of significant importance.

The proposed system combines automatic voltage protection with real-time energy monitoring to deliver a comprehensive solution for both residential and industrial applications. By continuously evaluating power factor and providing real-time feedback, **WattGuard** enables users to make data-driven decisions regarding load management and energy utilization. Ultimately, this system contributes to improved energy efficiency, reduced power losses, and enhanced reliability of electrical infrastructure.

1.1 Motivation

The growing dependence on electrical systems across residential, commercial, and industrial sectors has emphasized the demand for reliable and cost-effective solutions to protect electrical infrastructure. Voltage fluctuations, including under-voltage and over-voltage conditions, pose significant threats to equipment integrity, often resulting in system downtime, increased maintenance costs, and reduced operational efficiency. Moreover, inefficient power utilization—largely caused by a poor power factor—leads to excessive energy losses and elevated electricity expenses.

This project is driven by the need to overcome these challenges through an intelligent and integrated approach. By detecting voltage irregularities, monitoring energy consumption, and evaluating power factor in real time, the proposed system aims to ensure equipment protection while promoting energy efficiency. The inclusion of power factor analysis not only enhances overall system performance but also contributes to sustainable energy management practices across diverse applications.

1.2 Objectives

The primary objectives of this project are as follows:

1. To design and implement a reliable system capable of detecting and protecting electrical loads from under-voltage and over-voltage conditions, ensuring equipment safety and operational stability.

2. To develop a real-time monitoring framework for accurately measuring essential electrical parameters such as voltage, current, power factor, and total energy consumption.

3. To compute and analyze the power factor to evaluate system efficiency and provide actionable insights that enable users to optimize energy utilization and minimize power losses.

4. To integrate an automatic control mechanism that responds to abnormal voltage or power factor conditions, ensuring timely protection and maintaining consistent system performance.

5. To establish a scalable and cost-effective platform adaptable for both residential and industrial applications, promoting sustainable and efficient power management practices.

1.3 Report Outline

This report is structured into the following chapters:

Chapter 2: Background and Preliminaries: This chapter provides the theoretical foundation and essential background knowledge required to understand the concepts and technologies utilized in the project. It discusses the principles of voltage protection systems, energy metering, and power factor measurement, as well as their integration into a unified electrical protection and monitoring framework. Additionally, it presents a review of related literature, summarizing previous research efforts and technological advancements relevant to the development of this system.

Chapter 3: Design and Implementation: This chapter details the overall design methodology and implementation process of the project. It explains the system architecture, including both hardware and software components, and describes the operation of individual modules such as the AC voltage control unit, voltage protection circuit, energy metering module, and power factor measurement system. The integration of sensors—such as current and voltage sensors—with the microcontroller is elaborated, along with discussions on programming, calibration, and system testing procedures used to validate performance.

Chapter 4: Conclusion: This chapter summarizes the key outcomes and findings of the project. It evaluates the system's effectiveness in meeting its design objectives, discusses the challenges encountered during development, and highlights the practical significance of the results. The chapter concludes with recommendations and directions for future enhancements to improve system performance and expand functionality.

Chapter 2
Background and Preliminaries

This chapter presents an overview of the theoretical framework and apparatus employed in the development of the project titled **“WattGuard – Automatic Power Factor Correction & Voltage Protection System.”** It provides detailed explanations of the fundamental concepts underpinning the system, a review of related works and existing technologies in the field, and a comprehensive description of the tools, components, and methodologies used in the design and implementation of the proposed system.

2.1 Background

The project focuses on the integration of voltage protection and energy monitoring systems to ensure electrical safety and optimize power utilization. A solid understanding of key concepts such as voltage protection, energy metering, and power factor is essential to appreciate the system’s objectives.

- **Under- and Over-Voltage Protection:** These mechanisms safeguard electrical equipment from harmful voltage deviations outside a specified safe operating range. Under-voltage occurs when the supply drops below the threshold, while over-voltage arises when it exceeds the safe limit. Such fluctuations can damage appliances, reduce efficiency, and shorten the lifespan of electrical devices.

- **Energy Metering:** This involves the precise measurement of electrical energy consumption over time, typically recorded in kilowatt-hours (kWh), using electronic or electromechanical meters. Real-time energy monitoring enables users to track consumption patterns and identify opportunities for optimization.

- **Power Factor:** Defined as the ratio of real power (watts) to apparent power (volt-amperes), power factor indicates the efficiency of energy usage in the system. A low power factor signifies wasted energy and higher operational costs, highlighting the need for effective power factor correction.

Building upon these principles, WattGuard integrates current and voltage sensing, zero-crossing detection, and microcontroller-based processing to provide real-time monitoring of voltage, current, energy consumption, and power factor. The system not only protects appliances from voltage anomalies but also promotes energy efficiency by offering actionable insights and facilitating optimal load management in both residential and industrial applications.

2.2 Existing Works

Several studies have addressed aspects relevant to voltage protection, energy metering, and power factor measurement. The following works provide context and insight for the development of the proposed system:

- P. Rama Mohan, N. Mallikarjuna, and K. Niteesh Kumar, "A Novel Over-Voltage and Under-Voltage Protecting System for Industrial and Domestic Applications," *International Journal of Innovative Science and Research Technology (IJISRT)*, Vol. 5, Issue 10, Oct. 2020. This article reviews protection methods that safeguard industrial and residential electrical systems from harmful voltage deviations.
- A. Al-Naib and B. A. Hamad, "A Cost-Effective Method for Power Factor Metering Systems," *International Journal of Electrical and Computer Engineering Systems (IJECES)*, Vol. 13, No. 5, pp. 409–415, July 2022. The work proposes an approach for accurate and low-cost power factor monitoring.
- Engr. Manzoor Ellahi, Prof. Dr. Suhail A. Qureshi, and Mustansir Iqbal, "Power Factor Monitoring and Load Management Using Smart Metering Techniques," *International Journal of Engineering Research & Technology (IJERT)*, Vol. 2, Issue 12, Dec. 2013. This study explores microcontroller-based systems for energy monitoring and power factor assessment.
- Kunal Kalal, Saubhik Jaiswal, Darshan Joshi, Malay Vagadiya, Dr. Sweta Shah, "Modern Technique for APFC of Single Phase Systems," *International Journal of Engineering Research & Technology (IJERT)*, Vol. 9, Issue 5, May 2020. The paper describes an automatic power factor correction technique for single-phase loads.
- Md. Mejbaul Haque, Md. Kamal Hossain, Md. Mortuza Ali, Md. Rafiqul Islam Sheikh, "Microcontroller Based Single Phase Digital Prepaid Energy Meter for Improved Metering and Billing System," *International Journal of Power Electronics and Drive Systems (IJPEDS)*, 2011. The work focuses on microcontroller-based energy metering and billing applications.
- Md. Ashiquzzaman, Nadia Afroze, Taufiq Md. Abdullah, "Design and Implementation of Wireless Digital Energy Meter using Microcontroller," *Global Journal of Research in Engineering*, Vol. 12, Issue 2, Feb. 2012. This study presents a wireless energy monitoring system with real-time measurement capabilities.
- Iloh J. P., Odigbo A. C., Mbachu C. B., "Internet of Things Concept for Improving Energy Metering in Power Distribution Networks," *Iconic Research and Engineering Journals*, Vol.

6, Issue 11, May 2023, pp. 632–641. The paper discusses IoT-enabled energy metering systems for modern power networks.

- J. Markow, "Microcontroller-Based Energy Metering using the AD7755," *Analog Devices Application Note / Article*, (Year not specified). The article demonstrates microcontroller-based energy measurement using precision ICs.

- Image M. M. Mohamed Mufassirin & Ahamed Lebbe Hanees, "Cost Effective Wireless Network Based Automated Energy Meter Monitoring System for Sri Lanka Perspective," *International Journal of Information Technology and Computer Science (IJITCS)*, Vol. 10, No. 5, May 2018, pp. 68–75. The study presents a wireless energy monitoring approach for residential applications.

- S. M. Sultan & C. P. Tso, "Power Effectiveness Factor: A Method for Evaluating Photovoltaic Enhancement Techniques," *Processes*, Vol. 13, No. 8, 2025, Art. 2532. This work investigates power factor measurement and efficiency evaluation techniques in renewable energy systems.

2.3 Description of Apparatus

This section lists and provides a detailed description of the apparatus used in this project. The apparatus are essential for building the protection and monitoring system.

(model, functions, specifications, and operating principle) of each apparatus used in your project. Photos taken from the website are not allowed. Pictures must be prepared by yourself.

2.3.1 ESP32

Model

ESP32 Devkit V1

Functions

The ESP32 acts as the central controller of the system. It acquires data from voltage and current sensors, processes the information in real time, and executes the logic required to detect voltage anomalies, calculate energy consumption, and evaluate the power factor.

Specifications

- Microcontroller: ESP32-WROOM-32

- Operating Voltage: 3.3V

- Input Voltage (recommended): 5V

- Digital I/O Pins: 30 (varies with module)
- Analog Input Pins: 18 (12-bit ADC)
- Flash Memory: 4MB
- Wi-Fi and Bluetooth: Integrated

Operating Principles

The ESP32 is programmed to read analog signals from sensors such as the ACS712 current sensor and the voltage divider network. It computes real-time voltage, current, energy, and power factor values. Additionally, its built-in Wi-Fi and Bluetooth capabilities allow future expansion for wireless monitoring and data logging.

2.3.2 ACS712 Current Sensor

Model

ACS712-5A

Functions

The ACS712 sensor measures the current flowing through a circuit by producing an analog voltage proportional to the sensed current. It is used to measure the load current in the system.

Specifications

- Measurement Range:-5A to +5A
- Sensitivity: 185 mV per ampere
- Supply Voltage: 5V
- Accuracy: ±1.5

Operating Principles

The ACS712 current sensor employs a Hall-effect element to detect the magnetic field produced by current flowing through a conductor. This magnetic field is converted into an analog voltage proportional to the current, which can then be measured and processed by the microcontroller.

2.3.3 SCT-013 Current Transformer

Model

SCT-013-000

Functions

The SCT-013is anon-invasive current transformer used to measure AC current. It provides a current signal proportional to the measured current, which is then fed to the Arduino for processing.

Specifications

- Measurement Range: 0 to 100A AC
- Output: 50mA for a 100A input
- Secondary Voltage: 1V
- Accuracy: ±1

Operating Principles

The SCT-013 uses a toroidal transformer design to sense the magnetic field produced by current flowing through a conductor. The output is a scaled current signal that is converted into a voltage using a burden resistor, which is then read by the ESP32.

2.3.4 20x4 LCD Display

Model

HD44780

Functions

The LCD display is used to show real-time data, including voltage, current, energy consumption, and power factor, allowing the user to easily monitor the system's performance.

Specifications

- Type: 20x4 character LCD
- Voltage: 5V
- Communication: Parallel
- Display: 20 columns x 4 rows

Operating Principles

The LCDdisplay operates using an HD44780 controller. It communicates with the ESP32 through a parallel data interface, displaying the processed data from the ESP32 to the user.

2.3.5 Transformer

Model

Custom-built step-down transformer (240V to 12V)

Functions

The transformer steps down the AC voltage from 240V to 12V AC. It ensures the system operates at a safer voltage level for processing with the ESP.

Specifications

- Input Voltage: 240V AC
- Output Voltage: 12V AC
- Power Rating: 50W

Operating Principles

The transformer works on the principle of electromagnetic induction to reduce the voltage from the primary (high voltage) coil to the secondary (low voltage) coil, providing the required voltage for the system.

2.3.6 5V Relay Module

Model

SRD-05VDC-SL-C

Functions

The 5V relay module is used to control high-voltage devices based on the low-voltage control signals from the ESP. It is employed to cut off power in case of an over-voltage or under-voltage condition.

Specifications

- Voltage Rating: 5V DC (coil voltage)
- Maximum Load: 10A at 250V AC, 30V DC
- Switching Type: SPDT

Operating Principles

When the ESP sends a control signal to the relay, it energizes the coil, causing the relay's switch to either open or close, thus controlling the high-voltage devices.

2.3.7 Choke (Current Controlling Inductor

Model

Custom-built choke

Functions

The choke is used for controlling current flow, filtering out high-frequency noise, and providing inductive reactance to manage AC current in the circuit

Specifications

- Inductance: 10 µH
- Current Rating: 5A

Operating Principles

The choke operates by providing inductive reactance to alternating current. It opposes sudden changes in current, which helps stabilize the current flow and prevent spikes.

2.3.8 LM358 Operational Amplifier

Model

LM358

Functions

The LM358 operational amplifier is used in various stages of the project, including as a comparator to detect zero-crossing points in the AC waveform for power factor calculations.

Specifications

- Dual Operational Amplifier
- Supply Voltage: 3V to 32V
- Output Voltage: Rail-to-Rail

Operating Principles

The LM358 works by amplifying small signals and comparing them to a reference voltage, useful in detecting the zero-crossing points of AC waveforms, which are crucial for calculating the phase difference between voltage and current.

2.3.9 CD4030BE XOR Gate IC

Model

CD4030BE

Functions

The CD4030BE is an XOR gate IC used in this project for implementing logic circuits required for zero-crossing detection and phase comparison

Specifications

- Type: Dual 2-input XOR gate
- Supply Voltage: 3V to 18V
- Propagation Delay: 20ns

Operating Principles

The XOR gate in the IC compares two input signals and produces a high output when only one of the inputs is high. This feature is used in the system to detect the phase difference between voltage and current signals.

2.4 Description of Simulation Tools

The project uses various simulation tools to model and test the system before hardware implemen tation. The following simulation tools were employed.

(version, purpose, key features, applications) of each simulation tool used in your project.

2.4.1 Proteus

Version

Proteus 8.13

Purpose

Proteus is used for circuit simulation and PCB design. It helps to simulate the working of electronic components and verify the functionality of the system before physical implementation.

Key Features

- Microcontroller simulation
- Support for analog and digital components
- PCB design and routing
- Real-time simulation of Arduino projects

Applications

Proteus was used to simulate the entire voltage protection and energy metering circuit, including the Arduino, sensors, and voltage detection components. This tool enabled testing of the logic and functionality before hardware implementation.

2.4.2 Arduino IDE

Version

Arduino IDE 2.3.6

Purpose

The Arduino Integrated Development Environment (IDE) is used to write, compile, and upload code to Arduino boards. It provides an easy-to-use interface for developing and debugging embedded system applications.

Key Features

- Support for various Arduino boards
- Code editor with syntax highlighting and auto-completion
- Built-in serial monitor for real-time data visualization
- Wide range of libraries for interfacing with sensors and modules - Cross-platform support (Windows, MacOS, Linux)

Applications

The Arduino IDE was utilized for developing the code for real-time energy metering, voltage pro tection, and power factor calculation. The IDE enabled rapid prototyping and debugging of the firmware for the microcontroller, ensuring seamless integration with sensors and external hardware.

2.4.3 EasyEDA

Version

EasyEDA Standard Version

Purpose

EasyEDA is an online design tool used for schematic capture, circuit simulation, and PCB layout. It allows users to collaboratively design and simulate circuits in a browser-based environment.

Key Features

- Intuitive schematic editor
- Circuit simulation using integrated SPICE engine

- PCB design with customizable layers and footprints
- Extensive component library with real-time pricing and sourcing options
- Cloud-based platform with collaboration features

Applications

EasyEDA was used to design the schematic and PCB layout for the voltage protection and energy metering system. The platform's cloud-based environment facilitated collaborative design and en sured the generation of error-free Gerber files for PCB fabrication

Chapter 3
Design and Implementation

This chapter presents a detailed overview of the design and implementation of the project titled "WattGuard – Automatic Power Factor Correction and Voltage Protection System." The objective of this chapter is to describe the system architecture, key components, and operational principles, while highlighting the practical challenges encountered during development. Additionally, this section examines the system's performance, compares it with existing solutions, discusses the project budget, and explores its potential social and industrial impact in terms of electrical safety and energy efficiency.

3.1 Project Architecture

The project architecture of WattGuard is designed to integrate voltage protection, energy monitoring, and power factor measurement into a unified system. The main components and their functions are as follows:

AC Voltage Control: The system manages the 240V AC supply from the RUET power line. This control ensures proper testing and verification of the project's functionality while regulating the main power source. It safeguards the operation of the system and prevents damage due to voltage irregularities.

Voltage Protection System: This core subsystem ensures that the connected load operates safely within specified voltage limits. It protects against short circuits and abnormal voltage conditions, thereby enhancing equipment longevity, reliability, and overall operational safety.

Energy Meter: The energy meter measures critical electrical parameters, including voltage, current, power, and energy consumption. Accurate real-time monitoring of these parameters allows for precise evaluation of electrical performance and informed load management.

Power Factor Measurement: Power factor monitoring assesses energy efficiency and overall power quality. By measuring the phase difference between voltage and current waveforms, the system helps reduce energy losses, optimize electrical system performance, and minimize operational costs. Accurate power factor measurement is essential for effective energy management and sustainable electricity usage.

This integrated architecture ensures load protection, precise energy measurement, and enhanced power quality monitoring, providing a reliable and efficient solution for both residential and industrial applications.

3.1.1 Circuit Diagram

The circuit diagram is a crucial aspect of the design as it illustrates how the components are interconnected to form a complete system. The diagram shows connections between the ESP32, sensors, relay, and other essential components.

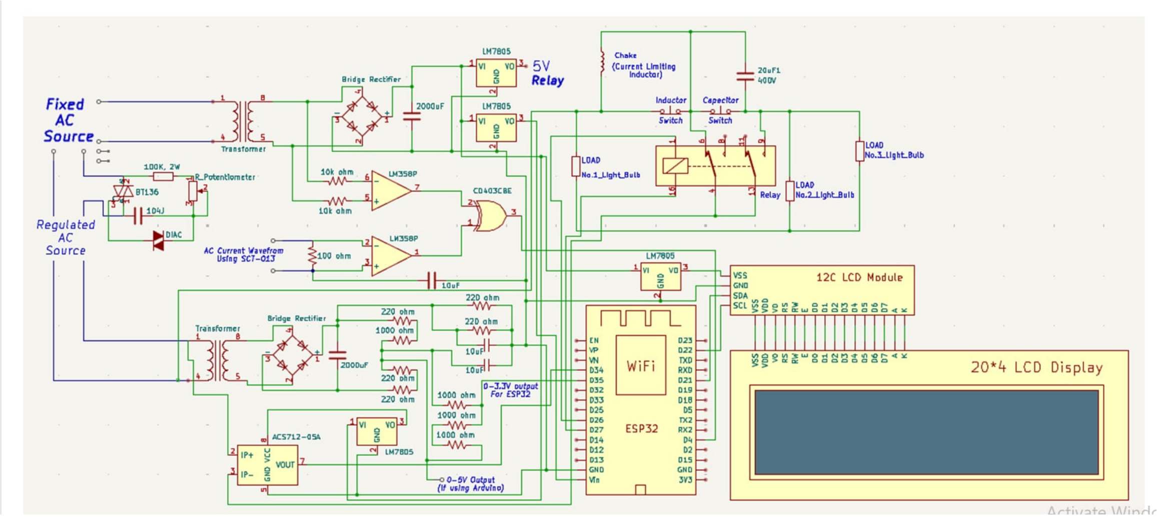

Figure 3.1: Final software setup of the system.

3.1.2 PCB Diagram

The PCB diagram is a crucial aspect of the design as it visually represents how the components are physically arranged and interconnected on the printed circuit board. It is based on the circuit diagram, which shows the logical connections between the ESP32, sensors, relay, and other essential components. The PCB diagram ensures proper placement of components, optimized routing of traces, and a compact layout for efficient and reliable system performance.

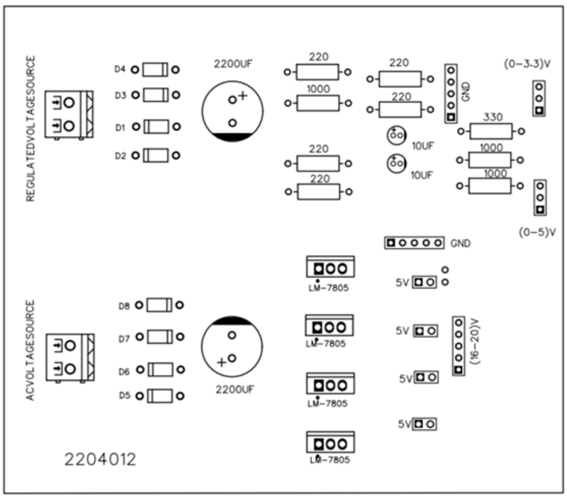

Figure 3.2: Top of Voltage Supply Circuit

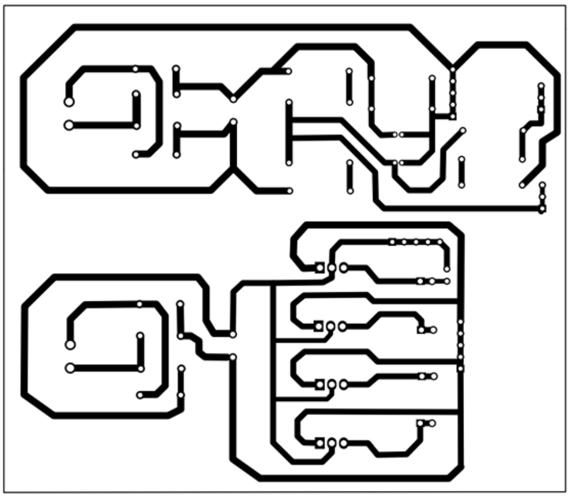

Figure 3.3: Bottom of Voltage Supply Circuit

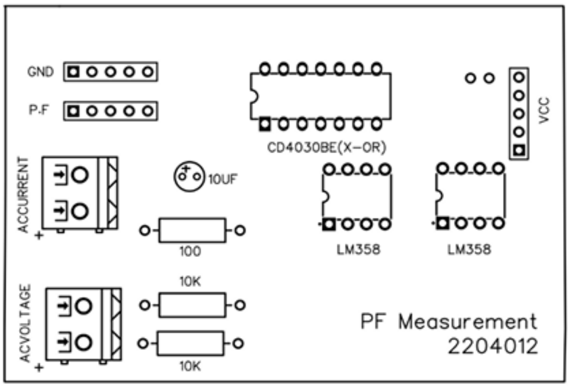

Figure 3.4: Top of PF Measurement Circuit

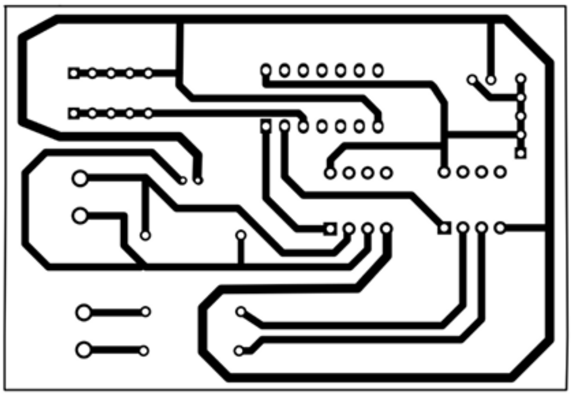

Figure 3.5: Bottom of PF Measurement Circuit

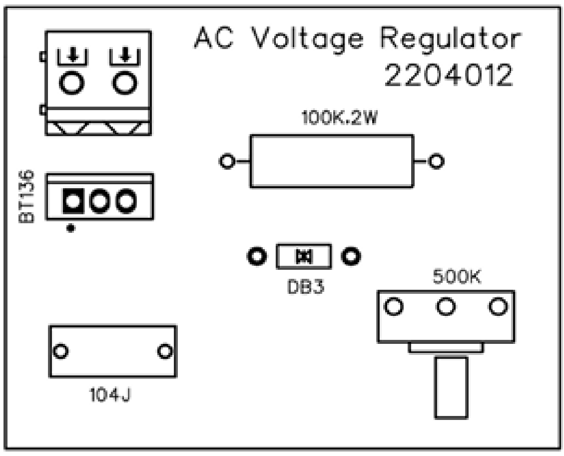

Figure 3.6: Top of Voltage Regulation Circuit

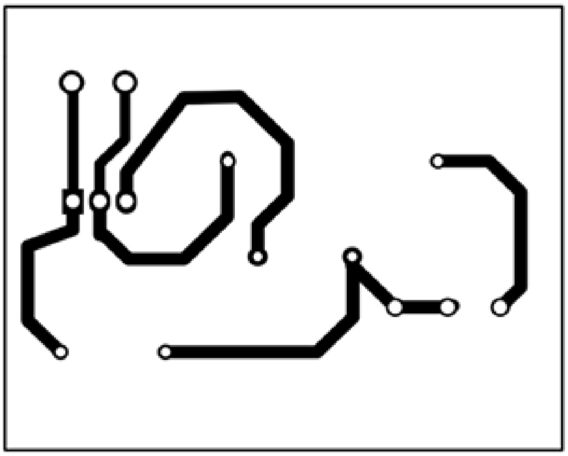

Figure 3.7: Bottom of Voltage Regulation Circuit

3.1.3 Hardware Diagram

The hardware diagram illustrates the physical setup of the components in the project.It includes key components such as the ESP32, LCD display, sensors, and the relay module.The arrangement of components is designed to ensure optimal system performance.Connections are organized for simplicity and ease of assembly.The hardware diagram helps visualize the integration of all com ponents in the project.This layout serves as a guide for assembling and troubleshooting the system effectively.The hardware diagram highlights the spatial relationship between components to min imize interference.Proper placement of components ensures efficient wiring and reduces potential errors.It provides a clear understanding of the physical connections and their functionality. The diagram is essential for replicating the project and maintaining consistency in assembly.

Figure 3.8: Working Hardware of the System

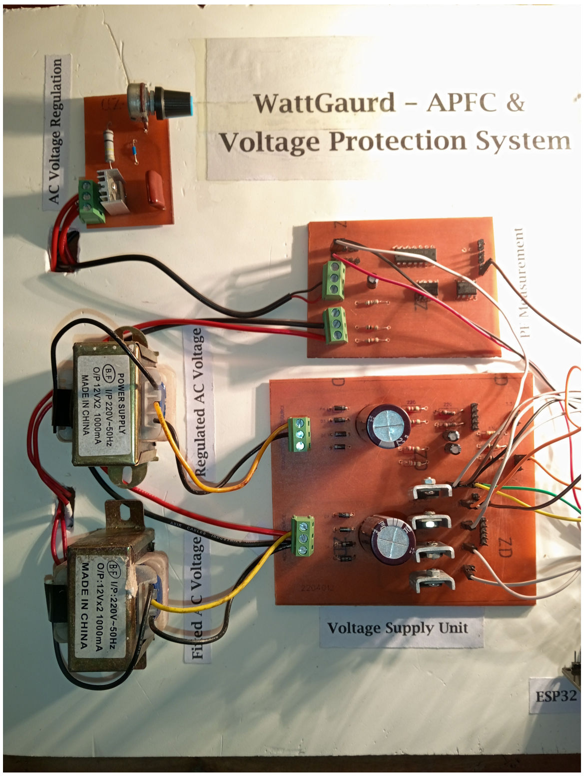

Figure 3.9: Working Hardware of the System

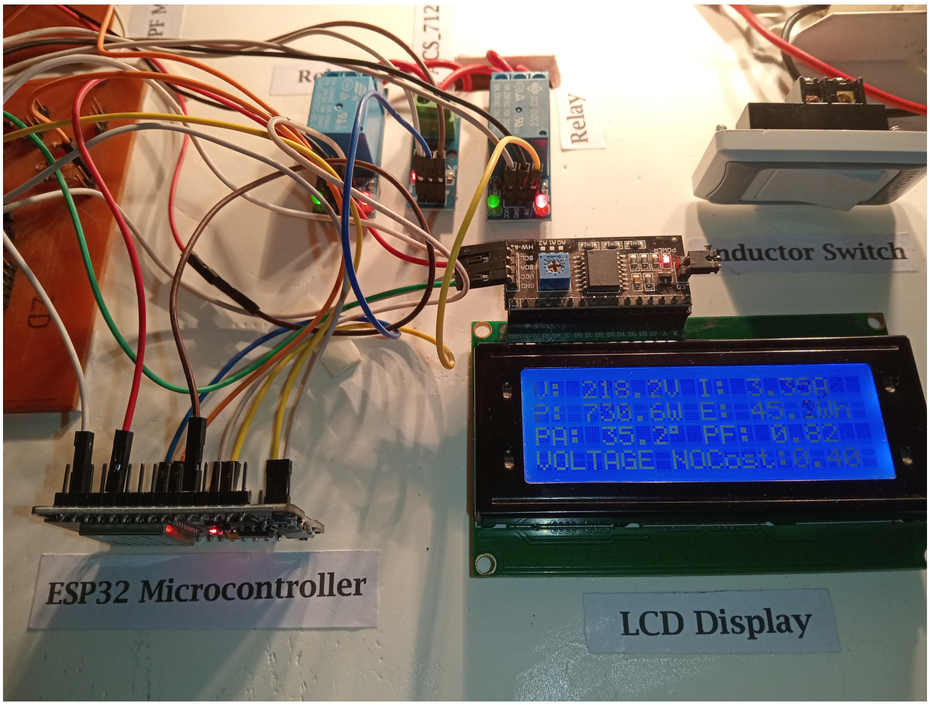

Figure 3.10: Working Hardware of the System

Figure 3.11: Working Hardware of the System

3.2 Operating Principle

The operating principle explains the step-by-step process of how the system functions to achieve the goals of under and over-voltage protection, as well as energy measurement with power factor calculation. The key steps in the operation are as follows:

Voltage and Current Measurement: The system utilizes a transformer-based voltage divider network and current sensors (ACS712) to measure the input voltage and current in real-time. For voltage measurement, the 0-240V AC input is stepped down using a transformer, rectified to DC using a bridge rectifier circuit, and further scaled down using a voltage divider network to produce a (0-5)V DCsignal compatible with the microcontroller's analog input. The microcontroller interprets this signal to calculate the corresponding 0-240V AC voltage. For current measurement, the ACS712 current sensor is directly connected to the load line, providing a proportional voltage output that represents the current flowing through the circuit.

Phase Difference Detection: The AC voltage waveform from the transformer and the current waveform from the SCT-013 current transformer are processed through the LM358 operational amplifier to generate pulse signals from their sinusoidal waveforms. These pulse signals are then fed into a 4030 XOR IC, which detects the intervals where the pulses do not overlap (i.e., one is HIGH while the other is LOW). The width of these intervals corresponds to the time difference between the voltage and current waveforms. This time difference is measured by the microcontroller, which calculates the phase difference based on the time measurement. We can calculate the phase angle manually by these equations:

$$X_L = 2\pi f L \tag{3.1}$$

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) \tag{3.2}$$

$$\cos(\phi) = \frac{R}{\sqrt{R^2 + X_L^2}} \tag{3.3}$$

$$P.E = \cos(\phi) = \frac{R}{\sqrt{R^2 + X_L^2}} \tag{3.4}$$

Equations (3.1)–(3.4) define the inductive reactance X_L, the phase angle ϕ (between voltage and current), the cosine of that phase angle, and the power factor (P.E) expressed as the cosine of the phase angle.

Protection Mechanism: The relay module controls the connection between the load and the power source. It disconnects the load when the voltage is outside the permissible range.

User Interface: The LCD display serves as the primary user interface, presenting real-time readings of voltage, current, energy consumption, and power factor. It receives data from the micro controller, which processes the measurements and executes the system's operational logic, ensuring accurate and updated information is displayed to the user.

3.3 Practical Applications

Basic application that can be implementable is:

1. **Households:** Protects appliances from damage caused by voltage fluctuations, ensuring safety and longevity.

2. **Industries:** Safeguards machinery from under-voltage and over-voltage issues, reducing down time and repair costs.

3. **Commercial Buildings:** Monitors energy usage and power factor, aiding in efficient energy management.

4. **Renewable Energy Systems:** Ensures stable operation of equipment in solar and wind power setups.

5. **Power Distribution:** Helps in monitoring and optimizing power flow in grids and substations.

6. **Educational Tools:** Demonstrates power factor and energy measurement concepts for students and researchers.

3.4 Technical Challenges/ Limitations

During the design and implementation, several challenges were encountered, including:

Sensor Calibration: The ACS712 current sensor required precise calibration to provide accu rate measurements, as temperature fluctuations could affect the readings.

Relay Control: Ensuring the proper switching of the relay module based on voltage thresholds involved handling the timing accurately to avoid damage to the load.

Noise Interference: The sensors and the Arduino faced interference from electrical noise. Im plementing filtering techniques, like using low-pass filters, helped to minimize this issue.

Proper Grouding: All separate sircuit have to have a common ground or oparetion will not syncronies.

These challenges were resolved by calibrating the sensors carefully, improving the relay control algorithm, and applying effective noise filtering techniques.

3.5 Social and Health Impact

The system provides a significant positive impact on society by offering an affordable and reliable solution for protecting electrical appliances from voltage fluctuations. By stabilizing voltage, it helps reduce the risk of damage to sensitive electrical devices. This protection extends the lifespan of appliances and reduces the frequency of costly repairs or replacements. In addition, the system improves energy efficiency by preventing the loss of power caused by voltage instability. The reduction in energy waste contributes to lower electricity bills for users. Safety is a key consideration in the system's design, ensuring proper insulation to protect users from electrical hazards. Grounding techniques are employed to ensure safety by directing excess electricity away from users and equip ment. The system also includes protective measures to handle high-voltage AC components safely. These features help prevent electrical fires or shocks, making it safer for household and industrial use. Overall, the system offers an effective way to safeguard appliances while promoting safer and more energy-efficient electricity use.

3.6 Budget of the Project

The estimated budget for the project is as follows-

ESP32: 450Tk
Current Sensors (ACS712, SCT-013): 350Tk
Relay Module: 100Tk
Transformer: 460Tk
LCD Display: 380Tk
Miscellaneous Components: 400Tk
Inductor (Chake): 250Tk
Bulbs & Holders : 270Tk
Normal & Jumper Wires : 50Tk
PCB: 250Tk

Total Estimated Budget: 3070Tk (Approximate)
Comparing the cost of the proposed system with commercial voltage protection systems and energy meters, the proposed solution is much more cost-effective, offering similar and more func tionality at a lower price.
As this was my first time undertaking this project, I had to test multiple circuits repeatedly and experiment with various components, many of which were not readily available in Rajshahi. Ad ditionally, some components turned out to be defective, resulting in higher expenses than initially estimated, with the total cost amounting to approximately 5000-6000 Tk

3.7 Comparison with Existing Works

Table 3.1: Comparison with existing works.

Existing Work	Proposed Work
Voltage Protection Systems Existing systems typically monitor voltage and disconnect the load when the voltage goes out of range. However, they do not measure power consumption or power factor.	**Under and Over Voltage Protection with Power Factor Measurement** The proposed system combines voltage protection with energy metering and power factor calculation, providing a comprehensive solution.
Energy Monitoring Systems Energy monitoring systems usually measure voltage and current for energy consumption calculation but lack real-time fault protection features.	**Integrated Energy Meter with Fault Protection** The proposed system integrates energy monitoring with under/over-voltage protection and power factor measurement for enhanced safety and efficiency.
Power Factor Measurement Traditional power factor measurement setups often require separate instruments and are not integrated into protection systems.	**Embedded Power Factor Measurement** The proposed design includes power factor measurement as part of a unified voltage protection and energy metering system.
Cost and Complexity Existing solutions are often more expensive and involve separate devices for protection, monitoring, and power factor measurement.	**Cost-Effective and Simplified Design** The proposed system consolidates multiple functionalities into a single, low-cost, and user-friendly design.

Chapter 4
Conclusions

4.1 Conclusions

The project titled **“WattGuard – Automatic Power Factor Correction and Voltage Protection System”** successfully developed a reliable, efficient, and cost-effective solution for monitoring, protecting, and analyzing electrical systems. By integrating multiple functionalities into a single system, the project achieved its objectives while maintaining simplicity and affordability. The key outcomes are summarized as follows:

- **Voltage and Current Monitoring:** The system accurately measured real-time voltage and current using dedicated sensors, ensuring precise and reliable data acquisition within operational limits.

- **Zero-Crossing Detection and Power Factor Measurement:** Through an op-amp-based zero-crossing detector, the system precisely measured the phase difference between voltage and current waveforms, enabling accurate power factor calculation for both resistive and inductive loads.

- **Under and Over Voltage Protection:** The relay module effectively disconnected the load under abnormal voltage conditions, protecting connected appliances and ensuring operational safety.

- **Energy Metering:** The system tracked energy consumption over time and displayed critical parameters such as voltage, current, power, energy, and power factor on an LCD interface, providing clear and actionable insights for users.

- **Cost-Effective Design:** By leveraging open-source hardware and straightforward circuitry, the project achieved robust performance at a fraction of the cost of commercially available solutions.

In conclusion, WattGuard demonstrates a scalable, practical, and robust design suitable for residential, commercial, and industrial applications. Its integrated approach ensures improved energy efficiency, enhanced safety, and reliable electrical performance. The successful implementation of this system provides a strong foundation for future advancements in electrical protection, energy monitoring, and smart power management.

4.2 Directions of Future Extensions

While the project met its primary goals, there are several opportunities for enhancement and further development. The following extensions are proposed for future work:

1. **Integration with IOT:** Incorporating IoT functionality into the system would enable remote monitoring and control via a web or mobile interface. Users could track real-time energy consumption and power factor data and receive alerts about voltage anomalies from anywhere.

2. **Enhanced Load Analysis:** Adding advanced analytics to classify and characterize different types of electrical loads (e.g., resistive, inductive, capacitive) would provide more detailed insights into energy usage and suggest ways to optimize power factor and efficiency.

3. **Data Logging and Visualization:** Introducing a data logging module with graphical visualization capabilities would allow users to analyze historical trends in energy usage, power factor variations, and voltage stability. This feature could aid in predictive maintenance and system optimization.

These enhancements would further improve the system's adaptability, usability, and functionality, making it a more comprehensive tool for modern electrical network management.

Appendix A

Simulation Codes

A.1 Codes for WattGaurd

```

#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include "EmonLib.h"
#include <math.h>

// ---------------------- LCD and Energy Monitor Initialization
    ----------------------
LiquidCrystal_I2C lcd(0x27, 20, 4);
EnergyMonitor emon1;

#define SDApin  21       // I2C SDA
#define SCLpin  22       // I2C SCL

// ---------------------- Pin Definitions ----------------------
const int xorPin = 4;         // XOR output (zero-cross detection)
const int relay1Pin = 26;     // Relay 1 (capacitor switching)
const int relay2Pin = 27;     // Relay 2 (inductor switching)
const int voltagePin = 35;    // Voltage sensor input
const int currentPin = 34;    // Current sensor input (ACS712)

// ---------------------- Constants ----------------------
const float ACS712_SENSITIVITY = 185.0;   // mV/A for ACS712 (5A version)
const float OFFSET_VOLTAGE = 2048.0;     // Midpoint of 12-bit ADC (4096/2)
const float referenceVoltage = 3.3;      // ESP32 ADC reference
const int numSamples = 500;              // Voltage averaging
const int sampleCount = 200;             // Current averaging
const int frequency = 50;                // Mains frequency (Hz)
const float PF_THRESHOLD_RELAY = 0.90;

// ---------------------- Variables ----------------------
float current = 0.0;
float RMSVoltage = 0.0;
float power = 0.0;
float energyWh = 0.0;          // energy in Wh
```

```
unsigned long lastTime = 0;
unsigned long lastUpdate = 0;

// ---------------------- Power Factor Stability ----------------------
float smoothedPhaseAngle = 0.0;   // EMA-filtered phase angle
float lastValidPF = 1.0;          // Fallback PF if signal lost
const float EMA_ALPHA = 0.1;      // Smoothing factor (0.1 = smoother, 0.5 =
     faster)

// ---------------------- Cost ----------------------
const float TARIFF = 8.95f; // currency per kWh
float total_cost = 0.0;
bool voltageTripped = false; // flag used to display TRIPPED / NORMAL

// ---------------------- Setup ----------------------
void setup() {
  Wire.begin(SDApin, SCLpin);
  lcd.init();
  lcd.backlight();
  Serial.begin(9600);

  pinMode(relay1Pin, OUTPUT);
  pinMode(relay2Pin, OUTPUT);
  pinMode(xorPin, INPUT);

  lcd.setCursor(0, 0);
  lcd.print("WattGuard System");
  lcd.setCursor(0, 1);
  lcd.print("Initializing...");
  Serial.println("WattGuard System Initializing...");
  delay(2500);
  lcd.clear();

  // -------- Main Display --------
  lcd.setCursor(0, 0);
  lcd.print("WattGaurd - APFC & ");
  lcd.setCursor(0, 1);
  lcd.print("Voltage Protection");
  lcd.setCursor(0, 3);
  lcd.print("Dhairjo - 2204012");
  delay(2500);
  lcd.clear();
```

```

  analogReadResolution(12);
  lastTime = millis();
}

// --------------------- Loop ---------------------
void loop() {
  // --- Current Measurement ---
  double sumOfSquares = 0;
  for (int i = 0; i < sampleCount; i++) {
    int rawValue = analogRead(currentPin);
    float adjustedValue = rawValue - OFFSET_VOLTAGE;
    sumOfSquares += adjustedValue * adjustedValue;
    delayMicroseconds(40);
  }
  float averageOfSquares = sumOfSquares / sampleCount;
  float rmsCurrent = sqrt(averageOfSquares);
  current = ((rmsCurrent * (referenceVoltage / 4096.0) * 1000.0) / 2) / ACS712_SENSITIVITY;

  if (!isfinite(current) || current < 0) current = 0.0;

  // --- Voltage Measurement ---
  long sum = 0;
  for (int i = 0; i < numSamples; i++) {
    sum += analogRead(voltagePin);
    delayMicroseconds(40);
  }
  float avgSensorValue = sum / (float)numSamples;
  // NOTE: Replace (185 / 1.57) with your actual divider/scaling after calibration.
  RMSVoltage = ((avgSensorValue / 4095.0) * referenceVoltage * (185.0f / 1.4f));

  if (!isfinite(RMSVoltage) || RMSVoltage < 0) RMSVoltage = 0.0;

  // --- Power and Energy Calculation ---
  power = RMSVoltage * current;
  unsigned long currentTime = millis();
  if (lastTime == 0) lastTime = currentTime;
  float deltaHours = (currentTime - lastTime) / 3600000.0f;
  energyWh += power * deltaHours;  // Wh = W * hours
```

```
lastTime = currentTime;

// --- Total Cost Measurement ---
total_cost = (energyWh / 1000.0f) * TARIFF; // convert Wh to kWh

// --- Robust Phase Angle & Power Factor from XOR ---
static unsigned long pulseBuffer[5] = {0};

// Read 5 pulse samples quickly
for (int i = 0; i < 5; i++) {
  pulseBuffer[i] = pulseIn(xorPin, HIGH, 20000); // 20ms timeout (safe for 50Hz)
  delayMicroseconds(100); // small gap between reads
}

// Simple bubble sort to find median
for (int i = 0; i < 4; i++) {
  for (int j = i + 1; j < 5; j++) {
    if (pulseBuffer[i] > pulseBuffer[j]) {
      unsigned long temp = pulseBuffer[i];
      pulseBuffer[i] = pulseBuffer[j];
      pulseBuffer[j] = temp;
    }
  }
}
unsigned long medianPulse = pulseBuffer[2]; // median value

float powerFactor = lastValidPF; // default: use last valid PF

// Validate pulse width: 100 s  to 8000 s  (      80   at 50Hz)
if (medianPulse > 100 && medianPulse < 8000) {
  float phaseAngleDegrees = (medianPulse * 360.0 * frequency) / 1000000.0;
  phaseAngleDegrees = constrain(phaseAngleDegrees, -80.0, 80.0); // realistic range

  // Apply exponential moving average (EMA)
  smoothedPhaseAngle = EMA_ALPHA * phaseAngleDegrees + (1 - EMA_ALPHA) * smoothedPhaseAngle;
  powerFactor = fabs(cos(smoothedPhaseAngle * PI / 180.0)); // use magnitude to avoid negative PF
  lastValidPF = powerFactor; // update fallback value
}
```

```
  // --- Display Phase Angle and Power Factor ---
  lcd.setCursor(0, 2);
  lcd.print("PA: ");
  lcd.print(smoothedPhaseAngle, 1);
  lcd.write(223); // Degree symbol
  lcd.setCursor(10, 2);
  lcd.print("PF: ");
  lcd.print(powerFactor, 2);
  lcd.print(" ");

  // --- LCD Update (every 100ms) ---
  if (millis() - lastUpdate >= 100) {
    lastUpdate = millis();

    lcd.setCursor(0, 0);
    lcd.print("V: ");
    lcd.print(RMSVoltage, 1);
    lcd.print("V  ");

    lcd.setCursor(10, 0);
    lcd.print("I: ");
    lcd.print(current, 2);
    lcd.print("A ");

    lcd.setCursor(0, 1);
    lcd.print("P: ");
    lcd.print(power, 1);
    lcd.print("W  ");

    lcd.setCursor(10, 1);
    lcd.print("E: ");
    if (energyWh >= 1000.0f) {
      lcd.print(energyWh / 1000.0f, 3);
      lcd.print("kWh");
    } else {
      lcd.print(energyWh, 1);
      lcd.print("Wh ");
    }
  }

  // --- Relay Control and Voltage Tripped Flag ---
```

```
  // voltage considered normal between 180..230 (adjust as needed)
  if (RMSVoltage >= 180.0 && RMSVoltage <= 230.0) {
    digitalWrite(relay1Pin, LOW);
    digitalWrite(relay2Pin, LOW);
    voltageTripped = false;
  } else {
    // outside normal range => tripped
    voltageTripped = true;
    // decide relays based on over/under if you still need physical actions:
    if (RMSVoltage > 230.0) {
      digitalWrite(relay1Pin, HIGH);
      digitalWrite(relay2Pin, LOW);
    } else { // below 180
      digitalWrite(relay1Pin, LOW);
      digitalWrite(relay2Pin, HIGH);
    }
  }

  // --- Bottom-line display: show VOLTAGE TRIPPED / NORMAL and Cost ---
  lcd.setCursor(0, 3);
  if (voltageTripped)
    lcd.print("VOLTAGE TRIPPED   ");
  else
    lcd.print("VOLTAGE NORMAL    ");

  lcd.setCursor(10, 3);
  // print cost on right side with 2 decimal places (no printf on this LCD lib)
  char costBuf[12];
  snprintf(costBuf, sizeof(costBuf), "Cost:%0.2f", total_cost);
  lcd.print(costBuf);

  // --- Serial debug (periodic)
  static unsigned long lastSerial = 0;
  if (millis() - lastSerial >= 1000) {
    lastSerial = millis();
    Serial.print("Vdc approx -> Vac=");
    Serial.print(RMSVoltage, 2);
    Serial.print("V, I=");
    Serial.print(current, 3);
    Serial.print("A, P=");
    Serial.print(power, 2);
```

```
    Serial.print("W, PF=");
    Serial.print(powerFactor, 3);
    Serial.print(", energyWh=");
    Serial.print(energyWh, 3);
    Serial.print(", cost=");
    Serial.println(total_cost, 2);
  }

  delay(10); // small delay to avoid hogging CPU
}
```

Listing A.1: Simulation Code for WattGaurd

Bibliography

[1] S. Ghosh, S. K. Sharma, and R. K. Aggarwal, "Power factor improvement in industrial loads using automatic power factor correction system," *International Journal of Electrical Power & Energy Systems*, vol. 105, pp. 345–352, 2019.

[2] R. Singh and P. K. Jain, "Design of an automatic power factor correction system using microcontroller," *Journal of Electrical Engineering & Technology*, vol. 15, no. 2, pp. 785–792, 2020.

[3] S. Patel and R. Patel, "Microcontroller-based voltage and current monitoring system for domestic and industrial applications," *International Journal of Engineering Research in Electronics and Communication*, vol. 5, no. 4, pp. 12–18, 2018.

[4] R. Maheshwari and K. K. Mehta, "Energy meter with real-time monitoring and load protection using arduino," *International Journal of Smart Grid and Renewable Energy*, vol. 6, no. 3, pp. 65–72, 2017.

[5] V. Khanna, "Current measurement using acs712 hall effect sensor and arduino," International Journal of Engineering Trends and Technology, pp. 11–16, 2019.

[6] MathWorks, "Simulink: Simulation and model-based design," https://www.mathworks.com/products/simulink.html, 2023.

[7] P. Gupta and A. Sharma, "Smart voltage monitoring and over/under voltage protection system for low voltage distribution networks," *International Journal of Electrical and Electronics Engineering Research*, vol. 11, no. 2, pp. 45–54, 2021.

[8] S. Sharma, R. Kumar, and P. Joshi, "Automatic power factor correction using capacitor banks and microcontroller-based control," *Journal of Power Electronics*, vol. 20, no. 4, pp. 1012–1021, 2020.

[9] L. Tan and M. Li, "Iot-based energy monitoring system for real-time power quality assessment," *IEEE Internet of Things Journal*, vol. 9, no. 7, pp. 5120–5130, 2022.